SORU KİTAPÇIĞI

Tarih:
2.05.2023

Sınavın,

Dersi:	BLM2502 - 1 Hesaplama Kuramı
Başlığı:	vize 1
Değerlendirme Türü:	Çoktan Seçmeli Sınav
Başlangıç Tarihi / Süresi:	25.04.2023 09:00 / 70 (Dk.)

Sorular,

1 / 24

A language is a set of strings.

10,00 Puan

A

True

B

False

2 / 24

The class of regular languages are closed under which of the following operations?

40,00 Puan

A

Union

B

Intersection

C

Complement

D

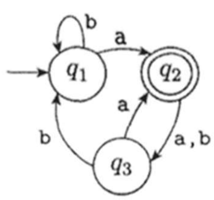
Concetanation

E

It is closed under all of other operations.

3 / 24

What sequence of states does the following automaton go through on input ����?



40,00 Puan

A

q_1 q_2 q_3 q_1 q_1

B

q_1 q_2 q_3 q_2 q_1

C

q_1 q_2 q_3 q_1 q_2

D

q_1 q_1 q_3 q_1 q_1

E

q_1 q_3 q_3 q_1 q_1

4 / 24

Every deterministic finite automata is also a non-deterministic finite automata.

10,00 Puan

A

True

B

False

5 / 24

A finite automaton can accept only one language, but a language can be accepted by more than one finite automaton.

10,00 Puan

A

True

B

False

6 / 24

$\delta(q_2, a) = q_1$ expression means that when the machine is in the state q_1 and input is a, the machine goes to state q_2.

10,00 Puan

A

True

B

False

7 / 24

If the language A is non-regular, then A is infinite.

10,00 Puan

A

True

B

False

8 / 24

Irregular languages are languages accepted by non-deterministic finite automata.

10,00 Puan

A

True

B

False

9 / 24

If A and B are regular languages, then A∘B is a context-free language. Here ∘ is concatenation operator.

10,00 Puan

A

True

B

False

10 / 24

Non-deterministic finite automata can accept languages that deterministic automata can not accept.

10,00 Puan

A

True

B

False

11 / 24

Languages that can be described by regular expressions are accepted by Finite Automata.

10,00 Puan

A

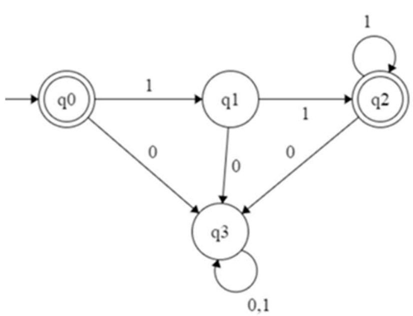
True

B

False

12 / 24

Which language does the finite automaton pictured in the figure below represents?



40,00 Puan

A

Complements a given bit pattern.

B

Finds 2’s complement of a given bit pattern.

C

Increments a given bit pattern by 1.

D

Changes the sign bit.

E

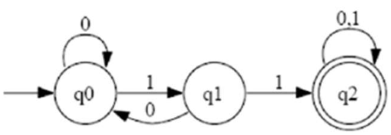
None of them.

13 / 24

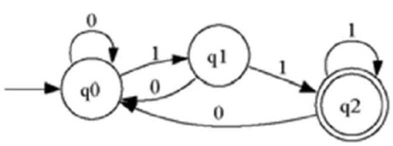
Which one of the following deterministic finite automata recognizes / accepts the language defined $L = \{w \in \Sigma^*$: each 1 in w is followed by a 0$\}$ over alphabet $\Sigma = \{0,1\}$?

40,00 Puan

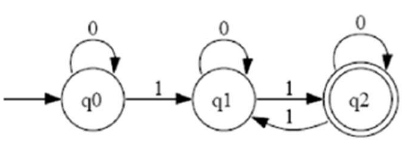
A



B



C



D

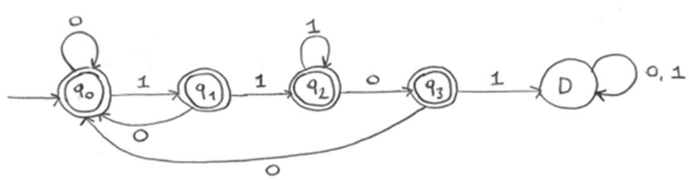
All of them

E

None of them

14 / 24

Which language does the following machine accept over the alphabet Σ={0,1}?



40,00 Puan

A

{w|w contains the substring 1101}

B

{w|w doesn' t contain the substring 1101}

C

{w|w contains the substring 110}

D

{w|w doesn' t contain the substring 110}

E

None of them.

15 / 24

Which language does the following machine accept over the alphabet Σ={0,1}?

40,00 Puan

A

{w|w contains the substring 0101}

B

{w|w doesn' t contain the substring 0101}

C

The empty string.

D

All strings except the empty string.

E

None of them.

16 / 24

Pumping lemma is used to show that a language is regular.

10,00 Puan

A

True

B

False

17 / 24

A context-free grammar is not ambiguous if it has more than one parsing tree at least for one string of its language.

10,00 Puan

A

True

B

False

18 / 24

Let the alphabet be Σ={a,b,c}. Which one of the following grammars is in Chomsky normal form?

40,00 Puan

A

$$S \to TTTa|a$$
$$T \to c$$

B

$$S \to TTa|a$$
$$T \to c$$

C

$$S \to TT|a$$
$$T \to c$$

D

$$S \to Ta|a$$
$$T \to c$$

E

$$S \to TT|ac$$
$$T \to c$$

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Following contex-free grammar is ambiguous.

$$S \to 1\,S\,1 \mid T$$
$$T \to 1\,X\,1 \mid X$$
$$X \to 0\,X\,0 \mid 1$$

10,00 Puan

A

True

B

False

20 / 24

Which is the language of the following context free grammar?

S→aSb | b

40,00 Puan

A $\{a^{n+1}b^{n}:\ \ n \geq o\}$

B $\left\{a^{n+1}b^{n+1}:\ \ n \geq o\right\}$

C $\{a^{n}b^{n}:\ \ n \geq o\}$

D $\left\{a^{n}b^{n+1}:\ \ n \geq o\right\}$

E None

21 / 24

Push-down automata transitions to only one different state for the same input.

10,00 Puan

A

True

B

False

22 / 24

(q_1,bbb,aaa$)≻(q_2,bb,aa$) tells that when a push-down automaton is in the state q_1 and gets the input b, it pops the symbol a from its stack, and transitions to state q_2.

10,00 Puan

A

True

B

False

23 / 24

If A has a regular expression, then A has a pushdown automata.

10,00 Puan

A

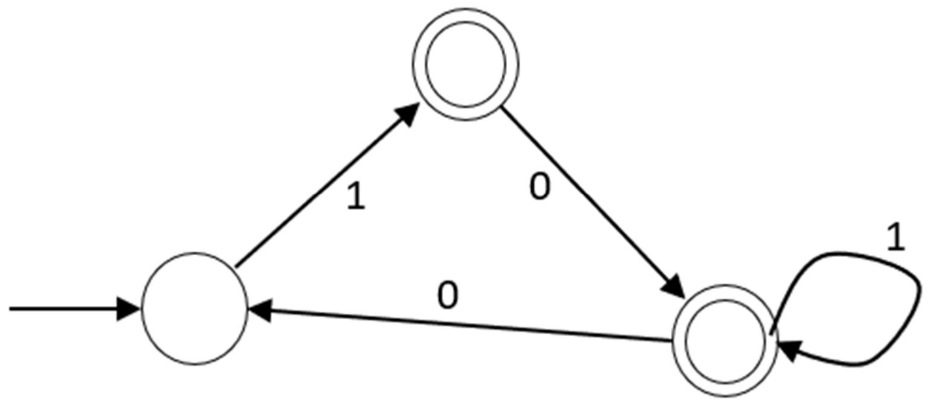
True

B

False

24 / 24

Which one of the following context-free grammars belongs to the automaton given inthe image?



40,00 Puan

A

S→1A
A→0B | ε
B→1B | 0S | ε

B

S→1A
A→0B
B→1B | 0S | ε

C

S→1A
A→0B | ε
B→1B | 0S

D

S→1A
A→0B
B→1B | 0S

E

None of them.